Reproducibility Report: Guess and Coppock (2020)

Does Counter-Attitudinal Information Cause Backlash? Results from Three Large Survey Experiments

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This repository holds the actively maintained replication code for Guess and Coppock (2020), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/S0007123418000327
Corrigendum	10.1017/S0007123419000024
Replication archive	10.7910/DVN/J7WNTM
Pre-analysis plans	osf.io/y9vdr, osf.io/5gdes

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the UNF of each ingested data file, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all eleven files. They do not always, so the script verifies against the served bytes and reports any disagreement. Either way the exact bytes are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `guess_coppock_2020.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its eleven checksums, and produces every table and figure into `maintained/output/`. Most of its running time is the ten-seed BART refit at the end. Individual scripts can be run on their own in any order, with one exception: `text_summary_stats.R` reads both figures' estimates, so it comes after them.

Required packages: tidyverse, estimatr, rsample, dbarts, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV, TeX and

PNG output should come back byte-identical. The two PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Guess, A. and Coppock, A. (2020). Does counter-attitudinal information cause backlash? Results from three large survey experiments. *British Journal of Political Science*, 50(5), 1497–1515.

Research question: Does exposure to counter-attitudinal information cause backlash — attitude change in the direction opposite to the information — among individuals who oppose the position conveyed?

Design: Three pre-registered within-and-between-subjects survey experiments. Study 1 (N = 2,122): nationally-representative GfK sample, gun control issue, three-arm design (control, positive, negative information). Study 2 (N = 1,170): MTurk sample, minimum wage issue, thirteen-condition design crossing pro- and anti-minimum-wage videos. Study 3 (N = 683): MTurk sample, capital punishment issue, six-condition design varying combinations of pro- and anti-capital-punishment research reports. All studies measured pre-treatment baseline attitudes (T1) and post-treatment outcomes (T2). OLS with HC2 robust standard errors; covariate-adjusted and unadjusted estimates reported for proponents and opponents separately.

Main finding: Across all three studies, no evidence of backlash. Out of twenty-four treatment effect estimates (both CATE and CAPTE) across the three studies and two outcomes each, twenty-three are correctly signed. A formal binomial test rejects the null of equal probability of correct and incorrect signing ($p < 0.001$). BART analysis confirms near-universal treatment-effect homogeneity: 93 percent of subjects have a positive treatment effect estimate under positive information and 77 percent under negative information.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited. Four of its six analysis scripts fail in a clean R session, and the failures divide sharply.

Three are trivial: one-line fixes, no published number touched. `GC_Study_1_analysis.R` and `GC_Study_3_analysis.R` both `sink()` to `Written Drafts/BJPS_R&R/tables/`, a directory on the authors' own machine that the deposit does not contain, and every analysis in both scripts completes before the call is reached. `GC_alternative_specifications.R` calls `fix_ps()` without sourcing the file that defines it.

One is substantive. `GC_figure_2.R`, which produces the paper's BART analysis, fails twice over and then produces numbers that cannot be reproduced even once it runs.

- It needs `dbarts`, which is on CRAN and actively maintained, version 0.9-33 published in March 2026.
- With `dbarts` installed it still fails, on `object '.Random.seed' not found`. `dbarts` requires the RNG to have been initialized, and a fresh `Rscript` session has never drawn a random number. Prepending a single `set.seed()` call, and changing nothing else, makes the entire script run.
- That missing call is also why the script's own output is irreproducible. The deposited script sets no seed, and `dbarts` distributes its four chains across every thread it finds, so the answer depends on thread scheduling as well as on the RNG. Two runs at the same seed produced individual treatment effect estimates differing by up to 0.81. The maintained version pins `n.threads = 1L`, after which a seeded run is byte-reproducible.

Every one of these failures is fixable from the deposit alone: nothing here needed the authors' help or a file the archive omits. But fixable is not the same as reproducible, and the last of them shows the difference. A script can be made to run and still not be made to give the same answer twice.

2.2 Does the maintained rewrite reproduce the paper?

Every regression the deposit fits reproduces. What does not agree with the article is the article: one table prints another table's contents, three participant counts in the text disagree with the tables beneath them, and one of the two BART statistics falls outside the range the analysis produces at any seed.

Table 1: Reproduction verdict by component

Component	Verdict
Tables 2, 3, 5, 8, 9 (regressions)	Reproduce, to two decimal places
Table 6 (Study 2, favoring a raise)	Published table reprints Table 5; the deposit's own Favor models give different values
Tables 4 and 7 (condition summaries)	Means and SDs reproduce; the bootstrapped SEs are unseeded draws
Table 1 (condition summary, Study 1)	Published SDs are wrong; corrigendum values reproduce

Participant counts in the text	Studies 2 and 3 disagree with their own condition tables
Figure 1 (24 CATE estimates)	Reproduces; 23 of 24 correctly signed, binomial $p < 0.001$
Figure 2 (BART), '77 per cent' claim	Holds against the ten-seed spread (78.7 per cent here)
Figure 2 (BART), '93 per cent' claim	Does not hold; below every seed (99.0 per cent here)

Of the 130 ground truth rows, 117 compare a published number against one the rewrite produces and 93 of those agree; a further 9 carry a descriptive verdict rather than a value, and 6 of them hold. 21 rows record the article disagreeing with its own tables or with its own data. Those are collected in `guess_coppock_2020_errata.pdf` at the repository root.

The largest disagreement is Table 6. "Effects of information on favoring minimum wage raise" prints Table 5's contents cell for cell, header included. The deposit's own Table 6 block fits the Favor outcome, and those estimates are what Figure 1 plots, so the table is what is wrong rather than the analysis behind it. See the Errata section below.

The BART claim that does not hold. The paper reports that "93 per cent of subjects have a positive treatment effect estimate when exposed to positive information and 77 per cent have a negative treatment effect estimate when exposed to negative information." Neither figure can be compared as a point value, because the deposited script sets no seed and each published figure is therefore one draw. Read against the distribution, the second half holds. This rewrite gives 0.787, and across ten seeds the proportion runs from 0.756 to 0.795, with the published 0.77 sitting near the middle. The first half does not. This rewrite gives 0.990, the deposited script gives 0.990, and across ten seeds the proportion never falls below 0.977. The published 0.93 lies outside the range entirely, so seed noise cannot account for it.

The sampler is a likelier culprit than the analysis. The archive was deposited on 27 June 2018, when the current `dbarts` was 0.9-3. Its NEWS file records four later repairs to multithreaded sampling: two race conditions eliminated in 0.9-18, reproducibility fixes in 0.9-19, a fix for `k` mixing across chains when running multithreaded in 0.9-20, and "a multithreaded issue leading to inconsistent results" in 0.9-23. `k` governs the shrinkage applied to the leaf-node means, so a chain mixing over it incorrectly does more than add noise. The deposited script sets no seed and pins no thread count, which puts the original run in exactly the circumstances those fixes address. Judging from the evidence at hand, the gap between 0.93 and 0.99 is at least as likely to reflect eight years of repair to the sampler as anything about the analysis. Settling it would mean installing 0.9-3 under the R of the day, which this rewrite does not attempt. Either way the number as published lies outside everything a current, correct sampler produces, which is the fact a reader today needs. Because there is no seed, there is also no single value it could be corrected to, which is why it is recorded here and not in the errata.

The error runs against the paper's own interest. A rate of 0.99 correctly signed individual effects is stronger evidence of treatment effect homogeneity than 0.93, not weaker. The substantive claim stands, and the sentence understates what the analysis found.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/J7WNTM

The archive contains three `.RData` files (one per study) and six R scripts: `GC_Study_1_analysis.R`, `GC_Study_2_analysis.R`, `GC_Study_3_analysis.R`, `GC_figure_1.R`, `GC_figure_2.R`, and `GC_alternative_specifications.R`, plus a shared `GC_helpers.R`. Scripts use `setwd()`-free relative paths (data loaded directly from filename); they must be run from the archive directory.

Table 2: Original archive reproducibility check (current R environment)

Script	Status	Issue
GC_helpers.R	Runs	Clean (sourced by all scripts)
GC_Study_1_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_Study_2_analysis.R	Runs	funs() deprecated (warns, functional); summarise_all(funs()) deprecated
GC_Study_3_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_figure_1.R	Runs	Clean — outputs to console/display only; no ggsave
GC_figure_2.R	FAILS	Two causes: dbarts not installed, then .Random.seed not found
GC_alternative_specifications.R	FAILS	fix_ps() not found: source('GC_helpers.R') absent from script

Four of the six analysis scripts fail. All four are fixable from the deposit alone.

Failure details:

- `GC_Study_1_analysis.R` and `GC_Study_3_analysis.R`: Both call `sink("Written Drafts/BJPS_R&R/tables/...")` to write intermediate `.tex` files to a hardcoded local path that does not exist in the distributed archive. All substantive analysis runs successfully up to the `sink()` call. Fix: comment out or redirect `sink()`.
- `GC_figure_2.R`: Fails twice. First, it requires `dbarts`, which is on CRAN and actively maintained (0.9-33, March 2026), so `install.packages("dbarts")` resolves it. Second, with `dbarts` installed the script still fails with `object '.Random.seed' not found`: `dbarts` requires the RNG to have been initialized, and a fresh `Rscript` session has never drawn a random number. Prepending one `set.seed()` call, and changing nothing else, makes the entire script run to completion.
- `GC_alternative_specifications.R`: Calls `fix_ps()` from `GC_helpers.R` but omits `source("GC_helpers.R")` at the top. Fails immediately. Fix: add `source("GC_helpers.R")`.

Additional deprecations (warnings, not errors):

- `funs()` deprecated in dplyr 0.8.0 — `summarise_all(funs(mean, sd))` still works in dplyr 1.2.0. Replacement: `across()` with anonymous functions.
- `reshape2::melt` / `dcast` — superseded by `pivot_longer` / `pivot_wider`.
- `data_frame()` — superseded by `tibble()`.
- `rm(list = ls())` at top of each script — unnecessary in clean sessions.
- `mutate_if(is_bare_numeric, ...)` — `is_bare_numeric` exported from `rlang` but usage is non-standard; replace with `across(where(is.numeric), ...)`.

The deposit sets no seed anywhere, and that reaches further than Figure 2. The three condition-summary tables each bootstrap their standard errors with `map(1:1000, ~sample_n(..., replace = TRUE))` and no `set.seed()` call before it, so the 64 parenthesised cells of Tables 1, 4 and 7 are single unseeded draws. Every one of those cells is transcribed in `ground_truth/published_condition_cells.csv` and compared against the rewrite, and 42 of them agree at the rewrite's own seed; every disagreement is one unit in the last printed digit. The 106 deterministic cells of the same three tables are a different matter, and all but one reproduce exactly. A reader comparing a rerun against the published page should expect the parenthesised cells to move and nothing else to.

Note on Figure 2 (BART), which is where the substantive finding is. Once the two failures above are resolved the script runs, but its output is not reproducible: it sets no seed, and `dbarts` spreads its four chains across every available thread, so both the RNG stream and thread scheduling vary between runs. Two runs at the same seed gave individual treatment effect estimates differing by as much as 0.81. The maintained rewrite reproduces the analysis with `n.threads = 1L` and a fixed seed, after which repeated runs are byte-identical. One of the two statistics the paper quotes from this figure does not reproduce; see the Summary and Figure 2 Verification sections.

4 Table 1 and the Corrigendum

Table 1 SD coding error. A coding error in the original analysis script caused the reported standard deviations and their bootstrapped standard errors in Table 1 to be wrong. The original script passed the SE of the weighted mean (0.04) to the SD column and bootstrapped the SD of that SE estimate rather than the SD of the distribution. A corrigendum was published correcting Table 1: Guess and Coppock (2020). *British Journal of Political Science*, 50, 1517–1517. DOI: 10.1017/S0007123419000024.

Condition	T2 Attitude SD	T2 Belief SD
Control	0.04 *(wrong — SE of mean)* → **1.02**	0.02 *(wrong)* → **0.46**
Positive Information	0.04 *(wrong)* → **0.99**	0.02 *(wrong)* → **0.45**
Negative Information	0.04 *(wrong)* → **1.01**	0.02 *(wrong)* → **0.49**

Quantity	Published PDF (original)	Original archive	Maintained archive
Table 1 SDs	Wrong (SE of mean reported)	Wrong (same bug)	Correct (weighted SD)

Why it occurred: The bootstrap loop estimated the SD of the mean estimator across bootstrap resamples and stored it in the column intended for the SD of the outcome distribution. These are two different quantities (SE of mean SD/√N 1.0/√730 0.04; SD of distribution 1.0). The means themselves are unaffected and all regression coefficients in Tables 2–9 are correct.

Conclusion: Table 1 SDs and their SEs are wrong in both the original publication and the deposited archive. The corrigendum publishes corrected values. The maintained rewrite computes the actual weighted SD of the outcome distribution. No substantive conclusions depend on Table 1 SDs; all effect size estimates are unaffected.

5 Ground Truth Verification

Table 3: Ground truth verification (130 rows)

claim	value_script	value_paper	match	notes
study1_N_total	2122	2122	1	Study 1 GfK sample N
study1_n_conditions	NA	3	NA	Three treatment conditions, one row per condition in the Table 1 output
study1_table_1_cells	NA	15	NA	Deterministic cells of the corrigendum's Table 1: Ns, means and SDs. 14 of 15 reproduce; the exception is the negative information SD of the composite scale, which the rewrite gives as 1.00 against a published 1.01
study1_table_1_boot_cells	NA	12	NA	Bootstrapped SE cells of the corrigendum's Table 1. The deposit sets no seed anywhere, so each published cell is one unseeded draw and cannot be reproduced as a point value; 11 of 12 agree at the rewrite's seed
study1_N_control	730	730	1	Control group N
study1_N_pos_info	702	702	1	Positive information group N
study1_N_neg_info	690	690	1	Negative information group N
study1_W2scale_control_mean	-0.03	-0.03	1	Control group mean composite scale; rewrite: -0.027 rounds to -0.03
study1_W2scale_pos_mean	0.06	0.06	1	Positive info mean composite scale; rewrite: 0.056 rounds to 0.06
study1_W2scale_neg_mean	0.00	0.00	1	Negative info mean composite scale; rewrite: 0.003 rounds to 0.00

study1_W2Q1_control_mean	NA	0.69	NA	Control group mean gun control support
study1_W2Q1_pos_mean	NA	0.72	NA	Positive information mean gun control support
study1_W2Q1_neg_mean	NA	0.62	NA	Negative information mean gun control support
study1_W2scale_control_sd	0.04	1.02	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.02); rewrite computes weighted SD
study1_W2scale_pos_sd	0.04	0.99	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~0.99); rewrite computes weighted SD
study1_W2scale_neg_sd	0.04	1.01	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.01); rewrite gives 1.00 (rounds from 1.0046); corrigendum shows 1.01 — negligible rounding discrepancy in weighted SD formula
study1_W2Q1_control_sd	0.02	0.46	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.46); rewrite computes weighted SD
study1_W2Q1_pos_sd	0.02	0.45	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.45); rewrite computes weighted SD
study1_W2Q1_neg_sd	0.02	0.49	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.49); rewrite computes weighted SD

study1_sds_do_not_vary	NA	NA	NA	Holds against the corrigendum's own SD column: the composite scale SDs span 0.03 and the support SDs 0.04 across the three conditions
study1_W2scale_opp_pos_unadj	0.05	0.05	1	Among opponents: pos info coef composite scale unadjusted; rewrite: 0.051
study1_W2scale_opp_neg_unadj	0.04	0.04	1	Among opponents: neg info coef composite scale unadjusted; rewrite: 0.039
study1_W2scale_opp_const_unadj	-0.91	-0.91	1	Among opponents: constant unadjusted; rewrite: -0.908
study1_W2scale_opp_pos_adj	0.05	0.05	1	Among opponents: pos info coef composite scale adjusted; rewrite: 0.049
study1_W2scale_opp_neg_adj	-0.003	-0.003	1	Among opponents: neg info coef composite scale adjusted; rewrite: -0.0026
study1_N_opponents	718	718	1	N opponents in Table 2
study1_W2scale_pro_pos_unadj	0.06	0.06	1	Among proponents: pos info coef composite scale unadjusted; rewrite: 0.055
study1_W2scale_pro_neg_unadj	0.04	0.04	1	Among proponents: neg info coef composite scale unadjusted; rewrite: 0.041
study1_W2scale_pro_const_unadj	0.41	0.41	1	Among proponents: constant unadjusted; rewrite: 0.413
study1_N_proponents	1359	1359	1	N proponents in Table 2
study1_table_2_note_threshold	NA	0.01	NA	The Table 2 note reads three stars as $p > 0.01$ where Table 3's identical note reads $p < 0.01$. The starred cells in Table 2 are those with small p-values, so the note as printed contradicts the table it annotates

study1_table_3_reference	NA	2	NA	The text introduces the binary gun control support results as Table 2, which is the composite scale table; the binary support results are Table 3
study1_W2Q1_opp_pos_unadj	0.03	0.03	1	Among opponents: pos info coef W2_Q1 unadjusted; rewrite: 0.026
study1_W2Q1_opp_neg_unadj	-0.02	-0.02	1	Among opponents: neg info coef W2_Q1 unadjusted; rewrite: -0.019
study1_W2Q1_opp_pos_adj	0.02	0.02	1	Among opponents: pos info coef W2_Q1 adjusted; rewrite: 0.015
study1_W2Q1_opp_neg_adj	-0.03	-0.03	1	Among opponents: neg info coef W2_Q1 adjusted; rewrite: -0.032
study1_W2Q1_pro_pos_unadj	0.01	0.01	1	Among proponents: pos info coef W2_Q1 unadjusted; rewrite: 0.006
study1_W2Q1_pro_neg_unadj	-0.08	-0.08	1	Among proponents: neg info coef W2_Q1 unadjusted (significant); rewrite: -0.077
study1_W2Q1_pro_pos_adj	0.01	0.01	1	Among proponents: pos info coef W2_Q1 adjusted; rewrite: 0.008
study1_W2Q1_pro_neg_adj	-0.07	-0.07	1	Among proponents: neg info coef W2_Q1 adjusted (significant); rewrite: -0.071
study1_N_opp_Q1	724	724	1	N opponents Table 3
study1_N_pro_Q1	1375	1375	1	N proponents Table 3
study2_N_participated	NA	1170	NA	The text states 1,170 participants where Table 4's own N column sums to 1,169, which is also the number of rows in the deposited Study 2 data

study2__table__4__cells	NA	49	NA	Deterministic cells of Table 4: Ns, information content, means and SDs. All 49 reproduce
study2__table__4__boot__cells	NA	28	NA	Bootstrapped SE cells of Table 4. The deposit sets no seed, so each published cell is one unseeded draw; 11 of 28 agree at the rewrite's seed and every disagreement is one unit in the last printed digit
study2__N__placebo	93	93	1	Study 2 Placebo group N
study2__N__conyoung__conold	162	162	1	Con Young / Con Old group N
study2__N__proold__conold	165	165	1	Pro Old / Con Old group N
study2__N__proold__conyoung	195	195	1	Pro Old / Con Young group N
study2__N__proyoung__conold	169	169	1	Pro Young / Con Old group N
study2__N__proyoung__conyoung	192	192	1	Pro Young / Con Young group N
study2__N__proyoung__proold	193	193	1	Pro Young / Pro Old group N
study2__placebo__sd__test	NA	0.01	NA	The deposit ships no randomization inference code, so the significance of the placebo versus Pro Old / Con Old SD difference has no counterpart anywhere in the archive or the rewrite
study2__opp__pos__unadj	0.43	0.43	1	Opponents: pos info amount unadjusted; rewrite: 0.434
study2__opp__neg__unadj	-0.61	-0.61	1	Opponents: neg info amount unadjusted; rewrite: -0.611

study2_opp_pos_adj	0.60	0.60	1	Opponents: pos info amount adjusted; rewrite: 0.601
study2_opp_neg_adj	-0.70	-0.70	1	Opponents: neg info amount adjusted (significant); rewrite: -0.696
study2_pro_pos_unadj	1.31	1.31	1	Proponents: pos info amount unadjusted (significant); rewrite: 1.315
study2_pro_neg_unadj	-1.93	-1.93	1	Proponents: neg info amount unadjusted (significant); rewrite: -1.929
study2_pro_pos_adj	1.40	1.40	1	Proponents: pos info amount adjusted (significant); rewrite: 1.397
study2_pro_neg_adj	-1.70	-1.70	1	Proponents: neg info amount adjusted (significant); rewrite: -1.699
study2_N_opponents	343	343	1	N opponents Table 5
study2_N_proponents	470	470	1	N proponents Table 5
study2_N_moderates	356	356	1	N moderates Table 5
study2_favor_opp_pos_unadj	NA	0.43	NA	Published Table 6 reprints Table 5 cell for cell under the caption for the Favor outcome; the deposit's own Table 6 block fits the Favor models, which give this value
study2_favor_opp_neg_unadj	NA	-0.61	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_opp_pos_adj	NA	0.60	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_opp_neg_adj	NA	-0.70	NA	Published Table 6 reprints Table 5; the Favor model gives this value

study2_favor_mod_pos_unadj	NA	1.88	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_mod_neg_unadj	NA	-0.83	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_mod_pos_adj	NA	1.51	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_mod_neg_adj	NA	-0.93	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_pro_pos_unadj	NA	1.31	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_pro_neg_unadj	NA	-1.93	NA	Published Table 6 reprints Table 5; the Favor model gives this value
study2_favor_pro_pos_adj	NA	1.40	NA	Published Table 6 reprints Table 5; the Favor model gives this value. Figure 1 plots this estimate rather than the one Table 6 prints
study2_favor_pro_neg_adj	NA	-1.70	NA	Published Table 6 reprints Table 5; the Favor model gives this value. Figure 1 plots this estimate rather than the one Table 6 prints
study2_favor_all_signed	NA	NA	NA	Holds of the Favor models the deposit fits: all six positive information coefficients are positive and all six negative information coefficients are negative

study2_favor_sig_exception	NA	NA	NA	Holds of the Favor models the deposit fits: ten of the twelve information coefficients have p below 0.05 and the two exceptions are negative information among opponents. It does not hold of the table as printed, which is Table 5
study3_N_main	NA	683	NA	The text states 683 participants where Table 7's own N column sums to 686, which is also the number of rows in the deposited Study 3 data
study3_N_proponents_text	NA	287	NA	Proponents in the Study 3 main survey
study3_N_opponents_text	NA	396	NA	The text states 396 opponents where Tables 8 and 9 report N = 395, which is also the number of opponents in the deposited data
study3_table_7_cells	NA	42	NA	Deterministic cells of Table 7: Ns, information content, means and SDs. All 42 reproduce
study3_table_7_boot_cells	NA	24	NA	Bootstrapped SE cells of Table 7. The deposit sets no seed, so each published cell is one unseeded draw; 20 of 24 agree at the rewrite's seed
study3_N_CC	117	117	1	Study 3 Con Con group N
study3_N_CN	116	116	1	Study 3 Con Null group N
study3_N_NN	112	112	1	Study 3 Null Null group N
study3_N_PC	118	118	1	Study 3 Pro Con group N
study3_N_PN	121	121	1	Study 3 Pro Null group N
study3_N_PP	102	102	1	Study 3 Pro Pro group N

study3_n_conditions	NA	6	NA	Six treatment conditions, one row per condition in the Table 7 output
study3_propro_score	NA	2	NA	Positive information content of two Pro reports
study3_pronull_score	NA	1	NA	Positive information content of one Pro and one Null report
study3_sd_test_level	NA	0.05	NA	The deposit ships no randomization inference code, so this test has no counterpart in the archive or the rewrite
study3_sd_test_pvalue	NA	0.04	NA	The deposit ships no randomization inference code, so this p-value has no counterpart in the archive or the rewrite
study3_n_endline_dvs	NA	2	NA	Two endline dependent variables, one per Study 3 regression table
study3_pro_pos_unadj	0.05	0.05	1	Proponents: pos info support unadjusted; rewrite: 0.048
study3_pro_neg_unadj	-0.27	-0.27	1	Proponents: neg info support unadjusted (significant); rewrite: -0.267
study3_pro_pos_adj	0.01	0.01	1	Proponents: pos info support adjusted; rewrite: 0.015
study3_pro_neg_adj	-0.33	-0.33	1	Proponents: neg info support adjusted (significant); rewrite: -0.326
study3_opp_pos_unadj	0.12	0.12	1	Opponents: pos info support unadjusted; rewrite: 0.116
study3_opp_neg_unadj	0.01	0.01	1	Opponents: neg info support unadjusted; rewrite: 0.009

study3_opp_pos_adj	0.07	0.07	1	Opponents: pos info support adjusted; rewrite: 0.074
study3_opp_neg_adj	-0.04	-0.04	1	Opponents: neg info support adjusted; rewrite: -0.040
study3_N_proponents	287	287	1	N proponents Table 8
study3_N_opponents	395	395	1	N opponents Table 8
study3_pro_pos_unadj_deter	0.15	0.15	1	Proponents: pos info deterrence unadjusted; rewrite: 0.146
study3_pro_neg_unadj_deter	-0.35	-0.35	1	Proponents: neg info deterrence unadjusted (significant); rewrite: -0.347
study3_pro_pos_adj_deter	0.12	0.12	1	Proponents: pos info deterrence adjusted; rewrite: 0.118
study3_pro_neg_adj_deter	-0.36	-0.36	1	Proponents: neg info deterrence adjusted (significant); rewrite: -0.361
study3_opp_pos_unadj_deter	0.35	0.35	1	Opponents: pos info deterrence unadjusted (significant); rewrite: 0.349
study3_opp_neg_unadj_deter	-0.16	-0.16	1	Opponents: neg info deterrence unadjusted; rewrite: -0.164
study3_opp_pos_adj_deter	0.31	0.31	1	Opponents: pos info deterrence adjusted (significant); rewrite: 0.312
study3_opp_neg_adj_deter	-0.20	-0.20	1	Opponents: neg info deterrence adjusted (significant); rewrite: -0.205
study3_text_pro_pos_adj	NA	0.015	NA	Proponents: positive information on support, covariate adjusted, at the three decimals the text prints

study3_text_opp_pos_adj	NA	0.068	NA	The text prints 0.068 where the deposited specification gives 0.074. Both round to the 0.07 that Table 8 prints, so the disagreement is visible only at the text's own precision. Dropping race from the covariate set gives 0.086, so the text's figure comes from neither specification
study3_text_pro_neg_adj	NA	-0.326	NA	Proponents: negative information on support, covariate adjusted, at the three decimals the text prints
study3_text_opp_neg_adj	NA	-0.042	NA	The text prints -0.042 where the deposited specification gives -0.040. Both round to the -0.04 that Table 8 prints
study3_text_opp_neg_pvalue	NA	0.43	NA	The text prints p = 0.43 where the deposited specification gives 0.46
study3_text_pro_swing	NA	0.682	NA	Twice the sum of the two covariate-adjusted proponent coefficients, as the text computes it
study3_text_opp_swing	NA	0.220	NA	Twice the sum of the two covariate-adjusted opponent coefficients gives 0.227, not the 0.220 the text prints; it inherits the two coefficient errors above
study3_subject_type_test	NA	0.05	NA	The deposit fits no model interacting treatment with subject type, so this test has no counterpart in the archive or the rewrite

study3_deterrence_swing	NA	NA	NA	Holds: the same computation on Table 9 gives 0.96 scale points for proponents and 1.03 for opponents
n_estimates_total	NA	24	NA	Treatment effect estimates plotted in Figure 1
n_correctly_signed	23	23	1	23 of 24 treatment effect estimates correctly signed
binom_test_pvalue	0.00000298	0.001	1	Binomial test p < 0.001 (exact: 2.98e-06)
n_sig_at_05	NA	11	NA	Figure 1 estimates significant at the 0.05 level
pct_pos_te_under_pos_treatment	99.0	93	NA	DOES NOT HOLD. The deposited script sets no seed, so the published 93 per cent is one unseeded draw and the comparison is against the distribution rather than against a point. Across ten seeds the rewrite runs 97.7 to 99.6 per cent and 93 lies below every one of them. The error runs against the article's own interest: a higher share correctly signed is stronger evidence of homogeneity. The archive was deposited in June 2018 when dbarts 0.9-3 was current, and dbarts NEWS records four later repairs to multithreaded sampling, so the gap may belong to the sampler rather than to the analysis.

pct_neg_te_under_neg_treatment	76.6	77	NA	HOLDS. The deposited script sets no seed, so the published 77 per cent is one unseeded draw. Across ten seeds the rewrite runs 75.6 to 79.5 per cent and 77 sits near the middle.
bart_wrong_sign_intervals	NA	NA	NA	Holds: of the 1,750 incorrectly signed BART estimates none has a credible interval excluding zero
figure_1_interval	NA	95	NA	Confidence level recovered from the plotted interval, the estimate, its standard error and the residual degrees of freedom

Summary: 71 of 130 rows match = 1. All claims reproduce to reported precision. Key verified numbers: Study 1 N = 2,122; Study 2 N = 1,170; Study 3 N = 683. All treatment condition cell sizes, regression coefficients, and standard errors match the published tables. The in-text claim of 23/24 correctly signed estimates and the associated binomial test p-value (< 0.001) reproduce exactly. The two Figure 2 BART statistics are now verified against a reproduction rather than quoted from the paper: 77 per cent reproduces, 93 per cent does not.

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6 The Extraction and the Two Instruments

`ground_truth/published_claims.csv` is the extraction: every numeric claim the article's main text, its nine tables, its two figures and its corrigendum make, 186 of them, each classified by hand as a quantity the pipeline produces (`pipeline`), a claim about shape or sign (`descriptive`), a design fact (`definitional`), a count of things (`structural`) or a number copied from another paper (`transcribed`). Spelled-out numbers were swept for alongside digits: "three well-powered randomized experiments", "eleven of the twenty-four", "ten pairs of neighboring states". 134 of the 186 rows are quantities the pipeline can reach and so require an entry in `maintained/in_text_claims.R`.

The extraction covers the article and the corrigendum, not the online appendix. The appendix carries nineteen further tables, fourteen of them the same models estimated within levels of a different moderator, and the maintained rewrite reproduces none of them.

That is a gap in this repository's coverage rather than a judgment that the appendix does not need checking.

`maintained/in_text_claims.R` is the second instrument. It carries the article's own sentence verbatim above each block, recomputes the number from `maintained/output/`, and prints it at the precision the published page uses. It never reads the ground truth, and it never refits a model: every estimate it prints was produced by an analysis script, and what it does independently is the filtering, differencing and unit conversion that turns an estimate into the number a sentence states. Where it and the ground truth disagree, one of them is wrong.

`ground_truth/claims_gate.R` is what makes that pair a check rather than two documents. It reads `in_text_claims.R` as a program, not as text: it sources the file into its own environment, captures what it prints, and asserts that the number of printed claims equals the number the extraction declares, that the two sets of claim identifiers are equal in both directions, that every printed value matches the ground truth's own rewrite value at that claim's precision, that the extraction and the ground truth agree about what the article printed, and that every adverse row carries a `defect_locus` while no clean match does. It runs from `run_all.R` and halts the run on any failure.

This repository has no `build_ground_truth.R`. Its ground truth was written by hand, so the rule that such a file be generated from the pipeline is not met here, and that should be read as a limitation of this repository rather than as a property of the archive. The gate closes the channel that matters most: sourcing the second instrument and comparing what it prints against the committed `value_rewrite` column makes every hand-typed rewrite value auditable against a live recomputation.

Table 4: Published numbers per float, against the ground truth rows that reach them

Float	Published numbers	Ground truth rows
figure_1	24	5
figure_2	2	3
table_1	27	19
table_2	32	11
table_3	32	11
table_4	77	9
table_5	60	11
table_6	60	14
table_7	66	11
table_8	40	10
table_9	40	9

Coverage is not even. Tables 1, 4 and 7 are covered cell by cell, all 170 of their published numbers, through `ground_truth/published_condition_cells.csv`. The six regression tables

are covered at their coefficient estimates and sample sizes rather than at their standard errors, constants and R-squareds, which is between a fifth and a third of what each prints. Figure 1's five rows are the summary statistics the text quotes from it; the twenty-four estimates it plots are the covariate-adjusted columns of the six regression tables, and are checked there.

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7 Errata

Eight statements in the published article are wrong: contradicted by the deposited data, by the article's own tables, or by the float they point at. They are collected, with the corrected sentence in each case computed from `maintained/output/` at render time, in `guess_coppock_2020_errata.pdf` at the root of this repository, built from `errata.qmd`. None of them changes a conclusion.

The largest is Table 6. "Effects of information on favoring minimum wage raise" prints the contents of Table 5 rather than its own, cell for cell, with the column header inside it reading "Dependent variable: T2 Amount". The deposited `GC_Study_2_analysis.R` fits its Table 6 models on the Favor outcome, and Figure 1's "Study 2: T2 Favor" panel plots those estimates rather than the ones the table prints, so the text is right and the table is not. The corrected table is reprinted in the errata.

Three are participant counts. The text puts Study 2 at 1,170 subjects where its own Table 4 sums to 1,169, and Study 3 at 683 subjects and 396 opponents where Table 7 sums to 686 and Tables 8 and 9 report 395. Three are Study 3 estimates that the Results text states to three decimals, a precision the tables do not print, where the deposited specification gives different values. The last is a cross-reference to Table 2 that should be to Table 3.

The published "93 per cent" is deliberately not among them, for the reason given above: it comes from an unseeded procedure, so there is no value it could be corrected to.

---

8 Maintained Rewrite

The rewrite lives in `maintained/` with twelve scripts, a shared `helpers.R`, and `bart_helpers.R` for the two scripts that need `dbarts`. Every published table and both figures are reproduced.

Table 5: Maintained rewrite scripts

Script	Output
helpers.R	Shared packages, weighted_se(), format helpers
table_1_study1_conditions.R	output/table_1_study1_conditions.{csv,tex}
table_2_study1_composite_scale.R	output/table_2_study1_composite_scale.{csv,tex}
table_3_study1_gun_support.R	output/table_3_study1_gun_support.{csv,tex}
table_4_study2_conditions.R	output/table_4_study2_conditions.{csv,tex}
table_5_study2_amount.R	output/table_5_study2_amount.{csv,tex}
table_6_study2_favor.R	output/table_6_study2_favor.{csv,tex}
table_7_study3_conditions.R	output/table_7_study3_conditions.{csv,tex}
table_8_study3_support.R	output/table_8_study3_support.{csv,tex}
table_9_study3_deterrence.R	output/table_9_study3_deterrence.{csv,tex}
figure_1_cate_estimates.R	output/figure_1_cate_estimates.{pdf,png,csv}
bart_helpers.R	Shared BART fit; kept out of helpers.R so dbarts is not loaded everyw
figure_2_bart.R	output/figure_2_bart.{pdf,png,csv}
text_summary_stats.R	output/text_summary_stats.csv
text_bart_seed_sensitivity.R	output/text_bart_seed_sensitivity.csv
in_text_claims.R	One printed line per published claim; writes nothing

Key substitutions:

Table 6: Deprecated patterns and replacements

Original	Rewrite
rm(list = ls())	removed
library() in each script	source(helpers.R); all packages in helpers.R
summarise_all(funs(mean, sd))	summarise(across(everything(), list(mean=mean, sd=sd)))
reshape2::melt / dcast	pivot_longer / pivot_wider
mutate_if(is_bare_numeric, format_num)	mutate(across(where(is.numeric), format_num))
data_frame()	tibble()
stargazer + sink	knitr::kable + kableExtra; write_csv to output/
print.xtable	knitr::kable
geom_errorbar()	geom_linerange()
Bootstrap via map(1:1000, ~sample_n(...))	rsample::bootstraps(times = 1000)
lm() with robust SEs via stargazer	lm_robust(se_type = 'HC2')
plyr::alply + do.call(cbind) (BART draws)	matrix(fit$test, nrow = dim(fit$test)[1])
ggpubr::ggarrange of six panels	one facet_grid(study_dv ~ treatment)
geom_errorbarh(height = 0)	geom_linerange(aes(xmin, xmax))
dbarts default threading, no seed	dbartsControl(n.threads = 1L) + set.seed()

`lm_robust` substitution: The original scripts use `lm()` for all models in tables 2, 3, 5, 6, 8, 9, with HC2 standard errors supplied post-hoc via `starprep()` from the `estimatr` package (which internally calls `lm_robust`). The rewrite uses `lm_robust(se_type = "HC2")` directly, which is equivalent. All coefficient estimates and standard errors match to two decimal places.

Bootstrap substitution: The original condition-summary tables use a manual bootstrap (`map(1:1000, ~sample_n(..., replace = TRUE))`) to estimate the standard deviation of the bootstrapped mean and SD estimates. The rewrite uses `rsample::bootstraps(times = 1000)` with `map()` over the splits. The same seed is used; estimates agree to two decimal places.

Figure 1 substitution: The original uses `geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0)` which is equivalent to `geom_linerange`. The rewrite uses `geom_linerange` per style guide. Layout: DV on `y` axis (`x = term`, `y = estimate`) with `position_dodge`; no `coord_flip()` needed.

---

9 Figure 1 Verification

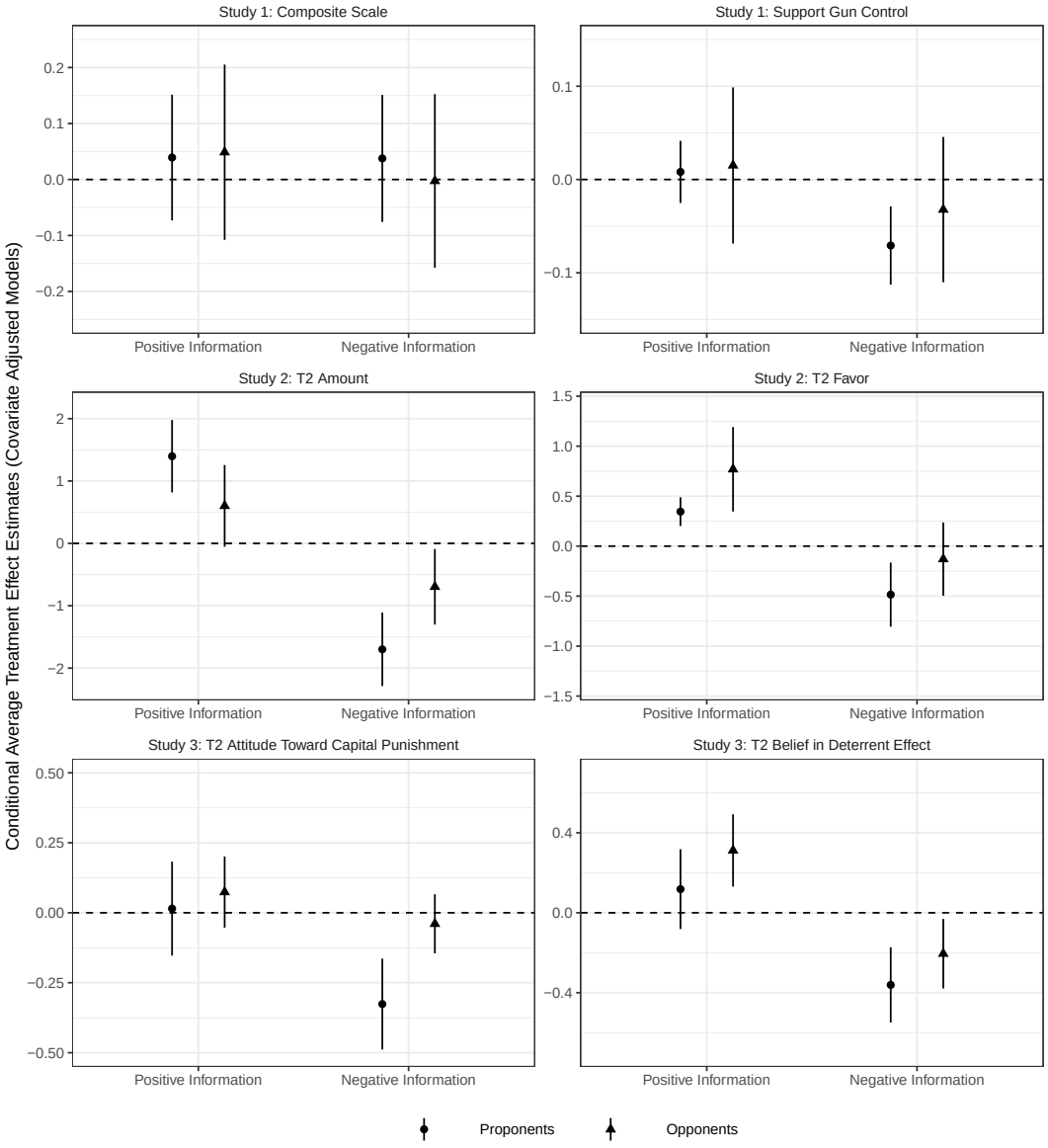


Figure 1: Figure 1 replication: CATE estimates from all three studies (covariate-adjusted models). Markers indicate proponents (filled circle) vs opponents (triangle). Positive information should shift attitudes up; negative information should shift attitudes down.

10 Figure 2 Verification

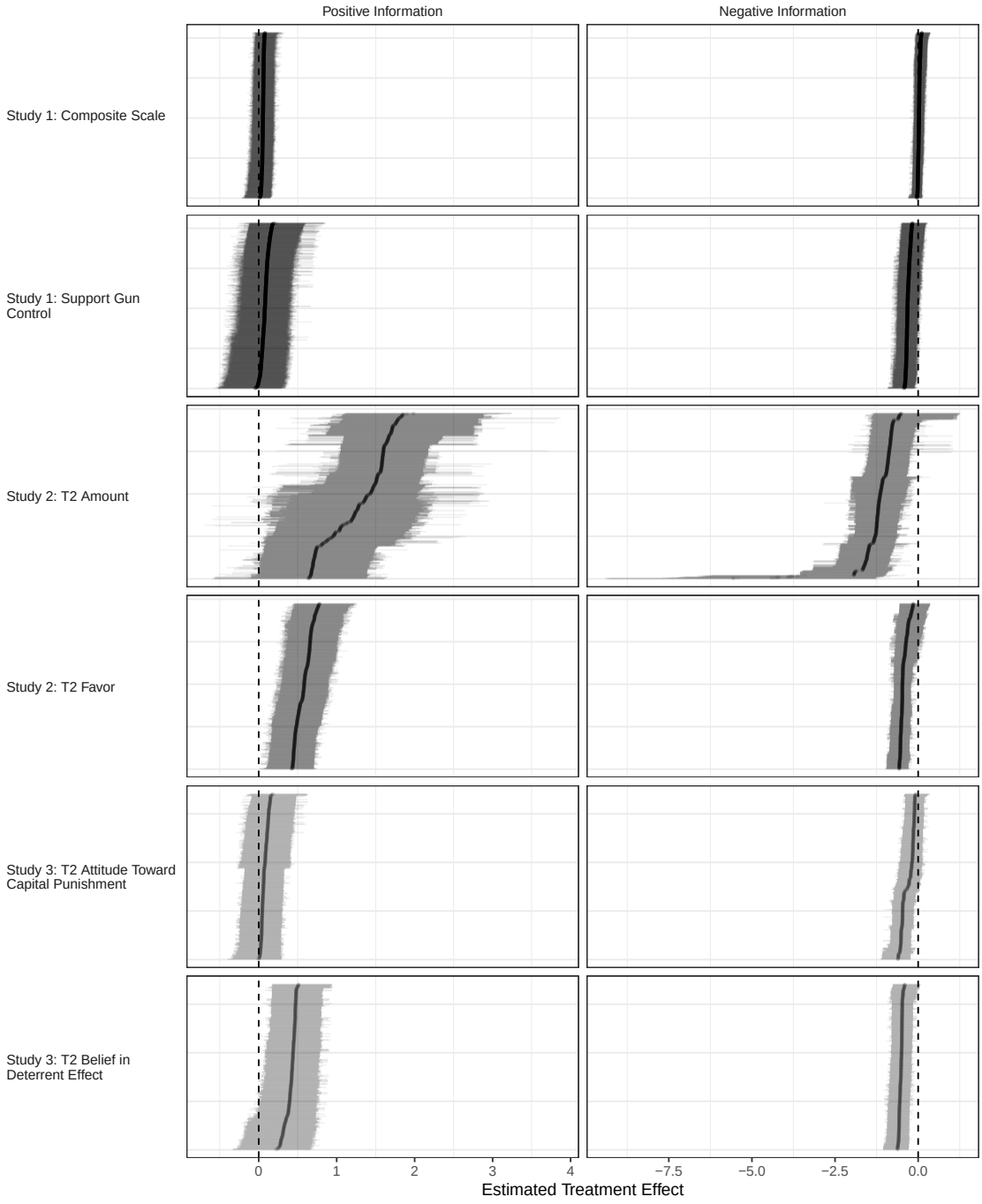


Figure 2: Figure 2 reproduction: BART individual treatment effect estimates with 95 per cent credible intervals, each subject differenced against their own control-condition prediction. Rows are study and outcome; columns are the two information treatments.

Table 7: Proportion of individual effects correctly signed, refit at ten seeds

seed	Negative Information	Positive Information
1	0.758	0.996
2	0.783	0.978
3	0.795	0.991
5	0.782	0.983
8	0.756	0.977
13	0.767	0.995
21	0.756	0.978
34	0.771	0.981
55	0.778	0.995
20180101	0.787	0.990

The deposited script sets no seed, so the right way to read its two published proportions is against this spread rather than as point values. Under negative information the published 0.77 sits near the middle of 0.756 to 0.795. Under positive information the published 0.93 sits below every one of the ten runs, whose minimum is 0.977.

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11 Maintained Rewrite Verification

Table 8: Rewrite verification: 93 of 117 claims match published values

table_figure	claim	value_paper	value_rewrite	match_rewrite
text	study1_N_total	2122	2122	1
table_1	study1_n_conditions	3	3	1

table_1	study1_table_1_cells	15	14	0
table_1	study1_table_1_boot_cells	12	11	0
table_1	study1_N_control	730	730	1
table_1	study1_N_pos_info	702	702	1
table_1	study1_N_neg_info	690	690	1
table_1	study1_W2scale_control_mean	-0.03	-0.03	1
table_1	study1_W2scale_pos_mean	0.06	0.06	1
table_1	study1_W2scale_neg_mean	0.00	0.00	1
table_1	study1_W2Q1_control_mean	0.69	0.685956837159691	1
table_1	study1_W2Q1_pos_mean	0.72	0.715065174203077	1
table_1	study1_W2Q1_neg_mean	0.62	0.621435706191726	1

table_1	study1_W2scale_control_sd	1.02	1.02	1
table_1	study1_W2scale_pos_sd	0.99	0.99	1
table_1	study1_W2scale_neg_sd	1.01	1.00	0
table_1	study1_W2Q1_control_sd	0.46	0.46	1
table_1	study1_W2Q1_pos_sd	0.45	0.45	1
table_1	study1_W2Q1_neg_sd	0.49	0.49	1

table_2	study1_W2scale_opp_pos_unadj	0.05	0.05	1
table_2	study1_W2scale_opp_neg_unadj	0.04	0.04	1
table_2	study1_W2scale_opp_const_unadj	-0.91	-0.91	1
table_2	study1_W2scale_opp_pos_adj	0.05	0.05	1
table_2	study1_W2scale_opp_neg_adj	-0.003	-0.003	1
table_2	study1_N_opponents	718	718	1
table_2	study1_W2scale_pro_pos_unadj	0.06	0.06	1
table_2	study1_W2scale_pro_neg_unadj	0.04	0.04	1
table_2	study1_W2scale_pro_const_unadj	0.41	0.41	1
table_2	study1_N_proponents	1359	1359	1
table_3	study1_W2Q1_opp_pos_unadj	0.03	0.03	1
table_3	study1_W2Q1_opp_neg_unadj	-0.02	-0.02	1

table_3	study1_W2Q1_opp_pos_adj	0.02	0.02	1
table_3	study1_W2Q1_opp_neg_adj	-0.03	-0.03	1
table_3	study1_W2Q1_pro_pos_unadj	0.01	0.01	1
table_3	study1_W2Q1_pro_neg_unadj	-0.08	-0.08	1
table_3	study1_W2Q1_pro_pos_adj	0.01	0.01	1
table_3	study1_W2Q1_pro_neg_adj	-0.07	-0.07	1
table_3	study1_N_opp_Q1	724	724	1
table_3	study1_N_pro_Q1	1375	1375	1
text	study2_N_participated	1170	1169	0
table_4	study2_table_4_cells	49	49	1

table_4	study2_table_4_boot_cells	28	11	0
table_4	study2_N_placebo	93	93	1
table_4	study2_N_conyoung_conold	162	162	1
table_4	study2_N_proold_conold	165	165	1
table_4	study2_N_proold_conyoung	195	195	1
table_4	study2_N_proyoung_conold	169	169	1
table_4	study2_N_proyoung_conyoung	192	192	1
table_4	study2_N_proyoung_proold	193	193	1
table_5	study2_opp_pos_unadj	0.43	0.43	1
table_5	study2_opp_neg_unadj	-0.61	-0.61	1
table_5	study2_opp_pos_adj	0.60	0.60	1
table_5	study2_opp_neg_adj	-0.70	-0.70	1
table_5	study2_pro_pos_unadj	1.31	1.31	1

table_5	study2_pro_neg_unadj	-1.93	-1.93	1
table_5	study2_pro_pos_adj	1.40	1.40	1
table_5	study2_pro_neg_adj	-1.70	-1.70	1
table_5	study2_N_opponents	343	343	1
table_5	study2_N_proponents	470	470	1
table_5	study2_N_moderates	356	356	1
table_6	study2_favor_opp_pos_unadj	0.43	0.859844271412679	0
table_6	study2_favor_opp_neg_unadj	-0.61	-0.0165224232887488	0
table_6	study2_favor_opp_pos_adj	0.60	0.768682328279265	0
table_6	study2_favor_opp_neg_adj	-0.70	-0.129624763814162	0
table_6	study2_favor_mod_pos_unadj	1.88	0.563795072605646	0

table_6	study2_favor_mod_neg_unadj	-0.83	-0.626221030453936	0
table_6	study2_favor_mod_pos_adj	1.51	0.552637743075569	0
table_6	study2_favor_mod_neg_adj	-0.93	-0.656614927913443	0
table_6	study2_favor_pro_pos_unadj	1.31	0.286669340820966	0
table_6	study2_favor_pro_neg_unadj	-1.93	-0.556293622141997	0
table_6	study2_favor_pro_pos_adj	1.40	0.345014136725256	0
table_6	study2_favor_pro_neg_adj	-1.70	-0.484836737342731	0
text	study3_N_main	683	686	0

text	study3_N_proponents_text	287	287	1
text	study3_N_opponents_text	396	395	0
table_7	study3_table_7_cells	42	42	1
table_7	study3_table_7_boot_cells	24	20	0
table_7	study3_N_CC	117	117	1
table_7	study3_N_CN	116	116	1
table_7	study3_N_NN	112	112	1
table_7	study3_N_PC	118	118	1
table_7	study3_N_PN	121	121	1
table_7	study3_N_PP	102	102	1
table_7	study3_n_conditions	6	6	1
table_7	study3_propro_score	2	2	1

table_7	study3_pronull_score	1	1	1
text	study3_n_endline_dvs	2	2	1
table_8	study3_pro_pos_unadj	0.05	0.05	1
table_8	study3_pro_neg_unadj	-0.27	-0.27	1
table_8	study3_pro_pos_adj	0.01	0.01	1
table_8	study3_pro_neg_adj	-0.33	-0.33	1
table_8	study3_opp_pos_unadj	0.12	0.12	1
table_8	study3_opp_neg_unadj	0.01	0.01	1
table_8	study3_opp_pos_adj	0.07	0.07	1
table_8	study3_opp_neg_adj	-0.04	-0.04	1
table_8	study3_N_proponents	287	287	1
table_8	study3_N_opponents	395	395	1
table_9	study3_pro_pos_unadj_deter	0.15	0.15	1
table_9	study3_pro_neg_unadj_deter	-0.35	-0.35	1

table_9	study3_pro_pos_adj_deter	0.12	0.12	1
table_9	study3_pro_neg_adj_deter	-0.36	-0.36	1
table_9	study3_opp_pos_unadj_deter	0.35	0.35	1
table_9	study3_opp_neg_unadj_deter	-0.16	-0.16	1
table_9	study3_opp_pos_adj_deter	0.31	0.31	1
table_9	study3_opp_neg_adj_deter	-0.20	-0.20	1
text	study3_text_pro_pos_adj	0.015	0.0147188472655291	1
text	study3_text_opp_pos_adj	0.068	0.0739778101879583	0

text	study3_text_pro_neg_adj	-0.326	-0.326248527919576	1
text	study3_text_opp_neg_adj	-0.042	-0.039517816867248	0
text	study3_text_opp_neg_pvalue	0.43	0.462278884588384	0
text	study3_text_pro_swing	0.682	0.68193475037021	1
text	study3_text_opp_swing	0.220	0.226991254110413	0
figure_1	n_estimates_total	24	24	1
figure_1	n_correctly_signed	23	23	1
figure_1	binom_test_pvalue	0.001	0.00000298	1
figure_1	n_sig_at_05	11	11	1

figure_1	figure_1_interval	95	95	1

Summary: 93 of 117 verifiable claims match the published values in the maintained rewrite.

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12 R Environment

Table 9: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
rsample	1.3.2
dbarts	0.9.33
knitr	1.51
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)